

PUBLISHING STRATEGIES AND ECONOMICS: FROM ARXIV TO NATURE

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August 27, 2025



This report summarizes business models in scholarly publishing, provides a practical scoring framework to rank venues by impact and cost, and compiles a compact comparison table of major platforms and publishers relevant to physics, information theory, computer science, distributed systems, and networking. A worked example of the value analysis is included in the appendix.

PRACTICAL NOTES

- Fees and policies change. Always verify article processing charges (APCs), licenses, and green open access (OA) rules on the specific journal page before submission.
- Read-and-publish (transformative) agreements can reduce author out-of-pocket APCs to \$0 for many publishers.
- Green OA often allows posting the accepted author manuscript (AAM) on arXiv or an institutional repository with a license specified by the publisher.
- Strategy: preprint on arXiv first, then target peer-reviewed venues that balance reach, prestige, rights, speed, and cost for your constraints.

BUSINESS MODELS IN SCHOLARLY PUBLISHING

READER-PAYS SUBSCRIPTIONS

Revenue primarily from library and individual subscriptions. Frequently paired with transformative deals that bundle reading plus OA publishing credits. Some journals still levy page or overlength charges; color charges are now uncommon.

AUTHOR-PAYS GOLD OA (APC-FUNDED)

Revenue from APCs. Hybrid journals offer an OA option alongside subscription access; fully OA journals rely entirely on APCs or sponsors. Waivers and discounts may be available (journal- and region-specific). OA licenses commonly include Creative Commons (for example, CC BY) .

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APC: Article Processing Charge	
OA: Open Access	
AAM: Accepted Author Manuscript	
CC BY: Creative Commons Attribution	

DIAMOND OR COMMUNITY-FUNDED OA

No fees for authors or readers. Costs are covered by consortia, libraries, foundations, and volunteer labor. Examples include overlay journals and community platforms.

SUBSCRIBE-TO-OPEN (S2O)

Libraries continue to subscribe; if enough participate each year, that year's volume becomes OA without APCs. If thresholds are not met, content remains gated for that cycle.

S2O: Subscribe-to-Open

SOCIETY CONFERENCES AND PROCEEDINGS

Revenue from registration fees, sponsorships, and exhibits. Proceedings are often free-to-read OA; the author's primary cost is registration and travel (with student grants or fee waivers sometimes available). The publisher's Version of Record (VOR) may be posted OA by the society; where only green OA is permitted, the AAM can typically be shared.

VOR: Version of Record

A SCORING FRAMEWORK FOR VENUE SELECTION

Let each venue be indexed by v . Define five dimensions and normalize them to the interval $[0, 1]$. Choose weights w_R, w_P, w_S, w_L to reflect your priorities (nonnegative, not necessarily summing to 1). Let cost $C_v > 0$ denote out-of-pocket cash required (APC or the minimum required registration).

DIMENSIONS

- Reach, R_v : visibility to the intended audience. Proxies include indexing coverage (for example, DBLP), Scopus, Web of Science, historical downloads, and whether the VOR or an AAM is openly available.
- Prestige, P_v : field-normalized reputation. Prefer SJR / SNIP or established field rankings over raw JIF.
- Speed, S_v : timeliness. For journals, median submission-to-accept and accept-to-publish times; for conferences, the fixed cycle from submission to notification.
- Rights, L_v : reuse freedom. CC BY (highest), then CC BY-NC or ND, then green OA with AAM sharing, then subscription-only with limited sharing.
- Cost, C_v : cash cost an author must pay. For conferences, include the cheapest legitimate registration path (student/remote if applicable).

DBLP: Computer Science Bibliography

SJR: SCImago Journal Rank

SNIP: Source Normalized Impact per Paper

JIF: Journal Impact Factor

AGGREGATE VALUE

Define the Value Index

$$\text{Value}(v) = \frac{w_R R_v + w_P P_v + w_S S_v + w_L L_v}{C_v}.$$

For quick looks, also compute Prestige-per-Dollar P_v/C_v and Reach-per-Dollar R_v/C_v . When $C_v = 0$ (for example, diamond OA), compare by the numerator $N = w_R R + w_P P + w_S S + w_L L$ and then break ties by editorial fit, indexing, and time-to-decision.

FIELD-SPECIFIC GUIDANCE

PHYSICS, INFORMATION THEORY, QUANTUM

Pair an arXiv preprint with a diamond OA or overlay venue (for example, community-funded physics journals) to maximize rights and reach at zero or minimal cost. Use prestigious society journals where green OA is permitted if APCs are unavailable.

COMPUTER SCIENCE, DISTRIBUTED SYSTEMS, NETWORKING

Top impact often flows through peer-reviewed conferences with OA proceedings (systems and networking). Budget for registration and travel; seek student grants where available. Journal extensions can follow for archival depth.

A FRUGAL PLAYBOOK

1. Preprint immediately on arXiv (choose a suitable license).
2. Target OA without APCs first (diamond, overlay, society-funded proceedings).
3. If aiming for a hybrid or flagship journal but funds are tight, use green OA: publish without paying APC and deposit the AAM per policy.
4. If you have an affiliation, check read-and-publish deals. If you are independent, look for waiver programs or S2O venues.
5. Prioritize venues with fast cycles if you need to establish precedence quickly.

WORKED MINI-EXAMPLES

OVERLAY OA PHYSICS JOURNAL

Reach high (OA + arXiv), prestige field-appropriate, rights strong (CC), speed moderate, cost near \$0. Value typically very high for a budget-

constrained author.

TOP SYSTEMS CONFERENCE PROCEEDINGS

Reach high (free-to-read proceedings), prestige very high, speed good (fixed cycle), rights adequate (archived OA proceedings), cost is registration/travel. Value high if travel support is secured.

HYBRID JOURNAL WITHOUT APC (GREEN OA)

Reach moderate (paywalled VOR, but AAM on arXiv), prestige varies by title, rights acceptable under green policy, cost \$0. Solid value when APC funds are not available.

FLAGSHIP GOLD OA

Reach and prestige very high, rights excellent (CC BY), speed variable, cost high. Prestige-per-dollar may be low for an unfunded author; use if funding or institutional coverage exists.

APPENDIX A: PUBLISHER OA POLICIES AND FEES

Platform	OA model(s)	Typical fees (APC or notes)
arXiv	Preprint server; author retains copyright; arXiv or CC license; no APC.	\$0.
Quantum (overlay)	Diamond OA physics overlay on arXiv; CC BY.	\$0 (check current site for any service fees).
SciPost	Diamond OA platform in physics; CC; consortial funding.	\$0.
APS (PR journals, PRX, PRX Quantum, PRResearch)	Hybrid and gold OA; CC licenses for OA; green self-archiving allowed.	PRL ~\$4k; PRX ~\$4.5k; PRX Quantum ~\$3.5k; PRResearch ~\$2.8k.
IOP Publishing	Hybrid and gold OA; CC licenses; many read-and-publish deals.	Varies by journal; see APC list.
IEEE (incl. IEEE Access)	Hybrid and gold OA; CC for OA articles.	Many hybrids ~\$2.6k; IEEE Access ~\$2.0k–\$2.2k.
ACM (journals, proceedings)	OA via ACM Open (institutional) or APC; authors retain rights for OA. Policy change in 2026.	If in ACM Open: \$0 to author; otherwise APC varies by venue.
USENIX (NSDI, OSDI, ATC)	Proceedings are free OA; no author APCs (registration separate).	\$0 APC for authors.
AIP Publishing	Hybrid and gold OA; CC for OA.	AIP Advances ~\$1.8k; others often ~\$2.7k.
Optica Publishing Group	Hybrid and gold OA; CC BY 4.0 available.	OA option ~\$2.2k–\$2.4k.
SIAM	Hybrid with strong green OA compliance; OA license varies by title.	APCs vary; green OA common.
Springer Nature (incl. Nature, npj, SpringerOpen)	Hybrid and gold OA; CC for OA.	Nature Comms ~\$6.9k; npj ~\$5.7k; Nature OA higher.
Elsevier	Hybrid and gold OA; generous green OA to arXiv (accepted MS).	Hybrids ~\$3k–\$4k; varies widely.
Wiley	Hybrid and gold OA; CC for OA; many transformative deals.	Varies by journal; often covered by deals.
Taylor & Francis	Hybrid (Open Select) and gold OA; CC for OA.	Varies; see cost finder.
Cambridge Univ. Press	Hybrid and gold OA; CC for OA.	Standard APC ~\$3.5k (varies).
Oxford Univ. Press	Hybrid and gold OA; CC for OA; self-archiving per journal.	Varies; many read-and-publish deals.
Royal Society (UK)	Hybrid plus RS Open Science (OA); moving to S2O model.	RSOS ~GBP 1.4k; S2O may remove APCs if thresholds met.
Royal Society of Chemistry	Hybrid and OA; CC for OA; many institutional deals.	Varies; often covered by agreements.
PLOS	OA-only; CC BY; waivers available.	PLOS ONE ~\$2.4k; others ~\$3k.
eLife	OA-only, preprint-centric; CC BY; waivable fee.	~\$3k publication fee.
Frontiers	OA-only; CC BY; APC per journal/category.	Roughly EUR 1.0k–2.7k+.
MDPI (Entropy, Quantum Reports)	OA-only; CC BY.	Entropy ~CHF 2,600; Quantum Reports ~CHF 1,400.

APPENDIX B: EXAMPLE VALUE ANALYSIS (SIX VENUES)

This example applies the scoring framework to six venues: Quantum, SciPost Physics, PRX Quantum, SIGCOMM, NSDI, and Nature Communications. Weights are

$$w_R = 0.35, \quad w_P = 0.35, \quad w_S = 0.15, \quad w_L = 0.15.$$

Values R, P, S, L are illustrative (0 to 1). Costs C are nominal out-of-pocket publishing/presentation costs (USD). For $C = 0$, we compare by the numerator $N = w_R R + w_P P + w_S S + w_L L$.

INPUTS AND SCORES

Venue	R	P	S	L	Cost C	Notes (waivers / hard costs)
Quantum (overlay)	0.80	0.70	0.60	1.00	\$0	Diamond/overlay OA; CC BY; no APC; check site for service-fee updates.
SciPost Physics	0.75	0.65	0.55	1.00	\$0	Diamond OA; community funded; no APC.
PRX Quantum	0.85	0.85	0.50	1.00	\$3,450	Gold OA; limited waivers; strong indexing and prestige.
SIGCOMM (ACM)	0.90	0.95	0.80	0.80	\$1,200	OA proceedings; at least one author registration typically required; travel not included.
NSDI (USENIX)	0.88	0.90	0.80	1.00	\$1,000	OA proceedings since 2008; author registration required; student grants available.
Nature Communications	0.95	0.90	0.45	1.00	\$6,990	Gold OA; waivers limited; many institutional read-and-publish deals.

COMPUTED NUMERATORS $N = w_R R + w_P P + w_S S + w_L L$

Venue	Numerator N	Value per \$1k = $1000N/C$	Prestige per \$1k = $1000P/C$
Quantum (overlay)	0.765	n/a ($C=0$; rank by N)	n/a
SciPost Physics	0.7225	n/a ($C=0$; rank by N)	n/a
PRX Quantum	0.8200	≈ 0.238	≈ 0.246
SIGCOMM (ACM)	0.8875	≈ 0.740	≈ 0.792
NSDI (USENIX)	0.8930	≈ 0.893	≈ 0.900
Nature Communications	0.8650	≈ 0.124	≈ 0.129

TAKEAWAYS

- By numerator N (ignoring cost), NSDI (0.893) and SIGCOMM (0.8875) lead, followed by Nature Communications (0.865) and PRX Quantum (0.82). Diamond OA venues Quantum (0.765) and SciPost (0.7225) are strong given zero cost.
- By Value per \$1k, NSDI (~ 0.893) and SIGCOMM (~ 0.740) dominate because registration costs are far lower than flagship APCs and proceedings are OA.
- For a budget-constrained researcher, a sequence of arXiv $\rightarrow$ USENIX/ACM systems venue $\rightarrow$ (optionally) a journal extension in a green-OA-friendly journal offers very high impact-per-dollar.
- In quantum/info-theory, Quantum and SciPost Physics deliver excellent rights and solid prestige at \$0 APC, with PRX-level venues adding additional prestige if funding is available.